

STANLEY CHEN

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Education

University of Waterloo

Systems Design Engineering

Sep. 2026 – Present

Waterloo, Ontario

Projects

CodeAtlas | *Python, GraphQL, GitHub Actions, JSON, Git*

July. 2026 – Present

- Built an automated synchronization system that retrieves accepted LeetCode submissions through the GraphQL API and generates organized, language-specific solution files and metadata.
- Engineered a modular synchronization pipeline with centralized language configuration, repository consistency checks, statistics generation, and timezone-aware metadata tracking.
- Designed a separate documentation layer that stores personal notes independently from submitted solutions, then injects language-appropriate notes into generated files while preserving the original solution code.
- Deployed scheduled and manual GitHub Actions workflows using repository secrets for credential management, automatically committing only when generated content changes.

Katharsis | *Python, Playwright, Chrome DevTools Protocol, Git*

Aug. 2026 – Present

- Built a browser automation tool using Playwright and the Chrome DevTools Protocol (CDP) to interact with an existing authenticated Chrome session without storing account credentials.
- Engineered detection and navigation logic for dynamically rendered web content, including message-type identification, ownership detection, and dynamically loaded conversation history.
- Developed safeguards and configurable limits for automating destructive browser actions while testing automation reliability against changing DOM structures.

Active Directory Home Lab | *Windows Server 2019, Active Directory, VirtualBox, Linux*

July. 2025 – Aug. 2025

- Built and troubleshooted a virtualized Windows domain environment with a Windows Server domain controller and Windows workstation to practice enterprise network administration.
- Configured Active Directory authentication, domain users, permissions, and Group Policy controls to simulate common IT administration workflows.
- Analyzed Windows Security Event Viewer logs and simulated authentication attempts to investigate failed logins and understand Windows security event logging.

Arduino Memory Game | *Arduino C++, TinkerCad, 3D Printing*

Jan. 2024 – June. 2024

- Designed, prototyped, and built an Arduino improvement of the "Simon Says" memory game with LEDs, push buttons, a buzzer, an LCD, and an IR remote system using TinkerCad and 3D printing.
- Implemented a difficulty function that adjusted pattern speed and a randomized pattern generation algorithm that increased sequence length after successful rounds.
- Developed custom tone functions and LCD user interface routines for game feedback and high score tracking.

Employment

Licensed 2 Learn

Jan. 2025 – June 2025

Mathematics Tutor

Toronto, Ontario

- Delivered 1:1 tutoring twice weekly to 5+ Grade 9–11 students, designing customized practice problems and lesson plans to strengthen algebra and functions fundamentals.
- Diagnosed gaps in student reasoning and adapted explanations using clear, practical examples to break down complex mathematical concepts.
- Delivered bilingual instruction in English and Mandarin to support ESL students.

Shoppers Drug Mart

Sep. 2024 – Jan. 2025

Pharmacy Assistant

Toronto, Ontario

- Supported pharmacists in processing prescriptions by entering patient and medication data into Healthwatch while maintaining strict confidentiality.
- Prepared prescriptions and handled inventory restocking using POS systems in a high-volume retail pharmacy environment.
- Assisted customers with prescription inquiries and provided English to Mandarin translation.

Skills

Programming: Python, C++, Java, Bash, JavaScript, HTML/CSS

Systems: Linux, Windows Server, Active Directory, Git, VirtualBox

Technologies: GitHub Actions, GraphQL, Playwright, Chrome DevTools Protocol (CDP), Arduino, TinkerCAD, CAD, 3D Printing

Spoken Languages: English (Native), Mandarin (Working), French (Basic)